SOCIAL MEDIA CYBERBULLYING DETECTION

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1. BUSINESS UNDERSTANDING

1.1 Overview

The kenyan digital landscape is one of the most vibrant in Africa, driven by a huge population of young, tech savvy and a high rate of mobile internet usage. The major platforms for social engagement, entertainment and business being FaceBook, YouTube, TikTok and X(formerly Twitter). Content creators, artists, vloggers and commentators commanding millions of comments daily.
Due to the anonymity and viral nature of these social media platforms, there is a developing culture if cyberbullying and online hostility. Unlike other Western countexts where comments fall under well known languages, Kenyan comment section is characterized by code -switching between Kiswahili, English and Sheng. Sheng is mostly being used to deliver insults or to "roast" individuals in the online space.

The current global cyberbulling moderation tools are primarily built in well-known languages like English, French, and a little bit of Swahili. Following the Kenyan linguistic complexity, there moderation tools fail to capture harmful sheng Phrases which leave kenyan content creators, public figures and online users vulnerable to online harrassmenr which in turn results to severe mental issues, reputation damages, suicidal ideation and digital exclusion.

Social media connects us, entertains us but sometimes it destroys us. Remember behind every trending post or hashtag, there is a real person with real feelings.

1.2 Problem Statement

There is a significant gap in the ability to automate systems that can detect and moderate cyberbulling within the linguistic Kenyan context.
Comments from Kenyan social media posts are based on sheng or code switching context which often bypass the existing moderation tools. The linguistic nuances of toxic content remain online, fueling a hostile digital environment. Content creators and brands are unable to moderate thousands of multilingual comments in real time, resulting in a toxic feedback loop that damages engagement and mental well-being.

1.3 Challenges

1. Linguistic Fluidity: Sheng is not standardized and a word might differ in meaning in different locations.
2. Code-mixed content: mixing English, swahili and sheng in one comment often confuse the existing NLP models
3. High Volume: Kenyan social media and digital space generate thousands of comments within hours which limits human moderation.
4. Contexual Sarcasm: While sheng is being used, deep contexual understanding is required to classfy comments whether "bullying", "hatespeech" rather than "Humor/Irony/Sarcism "
5. Data Scarcity: There is no publicly available high quality datasets for sheng specifically labelled for toxicity

1.3 Proposed Solution

We propose building an adavanced multilingual NLP classification system specifically tuned for the Kenyan linguistic landscape. A structured annotation through label studio for human annotation to ensure conssistent and high quality annotation of kenyan social media comments.
Each commenyt presented to the annotator will be labelled through four sequential steps:

1. Language Identification

2. Sentiment Classification
3. Major Category Classification
4. subcategory Classification

   This hierachical labeling will help analyze bothe the general and detailed information about the social media text

1. Language Identification: identify the primary language used in the comment The major language options include:

```
1. English
2. Swahili
3. Sheng
4. Mixed (Code-switching)
5. Plain Emoji
```

2. Sentiment Annotation : After identifying the language, the annotators will label the sentiment expessed in the comment.

   The sentiment categories include:

```
    1. Negative
    2. Positve
    3. Neutral
```

   Help understand the emotional tone of the message to detect harmful and constructive communications.

3. Major Category Classification : classify the comment into a main category that represents the overall nature of the message.

   The major categories include:

```
1. Constructive Criticism
2. Offensive Content
3. Cyberbullying
4. Hate Speech
5. Irony/Sarcasm
6. Misinformation
7. Other
```

4. Subategory Classification: each major category is linked to specific subcategories.

Subcategories Used in Data Annotation

Subcategories by Major Category

Constructive Criticism	Offensive	Cyberbullying	Hate Speech	Irony/Sarcasm	Misinformation	Others
[CC] Performance-based	[Offensive] Sex-based	[Bullying] Body shaming	[Hate Speech] Gender based	[Irony] Mock compliment	[Misinfo] Fabricated Content (Fake News)	[Other] Spam
[CC] Corrective clarification	[Offensive] Vulgar language (Matusi)	[Bullying] Direct Insult	[Hate Speech] Political	[Irony] Mock Laughter	[Misinfo] Misleading/Contextual Deception	[Other] General Others
[CC] Decision-criticism	[Offensive] Non-personal	[Bullying] Threats	[Hate Speech] Ethical/racial group (Tribalism)	[Irony] Rhetorical Question	[Misinfo] Parody	[Other] General Inquiry
[CC] Genuine Praise		[Bullying] Humiliation	[Hate Speech] Disability	[Irony] Hyperbolic understatement		
[CC] Supportive Speech		[Bullying] Exclusion tactics				
[CC] Genuine feedback		[Bullying] Sexual harassment				
		[Bullying] Trolling				
		[Bullying] Defamation				

1.5 Objectives

The goal is to develop a robust ML model that identifies and flags cyberbullying in Sheng, Swahili, and English with high accuracy.

• Scrape and curate specialized datasets of more than 10,000 comments from Kenyan YouTube vlogs.

• Use human annotators (Fluent Sheng speakers) to create high-quality labelled datasets

• Evaluating the model performance using F1-score, ensuring the model handles imbalanced data where bullying is less frequent than neutral comments.

• Build a prototype API or Dashboard that content creators can use to clean their comment sections automatically.

1.6 Success Criteria

This project will be considered successful if:

• The model achieves an F1-Score of 75% or higher across the cyberbullying and hate speech categories.

• The system successfully identifies Sheng-specific insults that standard English filters miss.

DATA UNDERSTANDING

2.1 Overview

To solve the problem identified in this project, we leveraged data from YoutubeTube comments from various Kenyan videos. The data will be scraped from YouTube using the YouTube Data API V3, an official and legal way of accessing public YouTube data for analysis. The project leverage the google-api-python-client interface with the comment Threads resource.
The Scripts maps the raw JSON responses from YouTube and converts them into Pandas dataframe
The data contains the following columns:

• Author: the public name of the commenter
• Text: raw comment content in plain text
• Likes: engagement metric
• Published at: Timestamp of the post.

Through this strategy, we managed to scrap over 10,000 comments from Kenyan Youtube Content.

2.2 Labelling

HUMAN DATA LABELLING AND ANNOTATION(Label Studio)

As compared to automated labeling methods, human annotators are better to interprate the nuances in the comments.

Human annotation ensure accurate detection of harmful content in categories such as Cyberbullying, hate speech, offensive language and Sarcasm . These categories require human judgement because the meaning often depends on context and intention.

- Human labeling helps ensure:
- Correct classification of comments
- Reliable sentiment detection
- Consistent labeling across categories
- High-quality training data for the NLP model.

2.3 Data Quality issues

Major data quality issues during the annotation process include;

1. A significant amount of data use non standard english hence annotator needs a local Kenyan Slang understanding to accurately identify the segments
2. class imbalance: Categories such as Contructive critism and Irony appear way more than other .
3. Certain names appear in toxic comments highlihting a need for annymization(masking). In data preparation to prevent the model from learninf "name_based" bias.

Data Loading and Inspection

In [ ]:
```
import sys
import os

sys.path.append(os.path.abspath("..."))
```

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

## Import Class Modules
from Data_prep.loader import DataLoader
from Data_prep.eda import EDA
```

Loading Dataset in a Dataframe

In [ ]:
```python
loader=DataLoader("final_data.csv")
data= loader.load_data()
data.head()
```

```
✅ Dataset loaded successfully from final_data.csv
```

Out[ ]:

	Unnamed: 0.2	Unnamed: 0	Unnamed: 0.1	annotation_id	annotator	author	category	clean_text
0	0	NaN	5028	3011	1	@Annemimi234	Irony/Sarcasm	Napenda hii couple but hapo kwa 29yrs nimekata...
1	1	NaN	5029	3012	1	@MKLMNMKLMNMKLMN	Others	rada zako wasingam
2	2	NaN	5030	3013	1	@raphaelmutinda8214	Constructive Criticism	Izo chwom mbaya ndio zetuu face_with_tears_of_...
3	3	NaN	5031	3014	1	@rhinoriders6060	Constructive Criticism	Nka shy grinning_face_with_sweat
4	4	NaN	5032	3015	1	@wanjawakariuki9079	Constructive Criticism	Ata kama ni dooh kuna shida zingine siwezi taka

Dropping Colums

In [ ]:

```
columns_to_drop= ["Unnamed: 0","Unnamed: 0.2","Unnamed: 0.1","annotation_id","annotator",
                  "author","clean_text","confidence_score","created_at","id",
                  "lead_time","predicted_label","updated_at"]
data=loader.drop_irrelevant_columns(columns_to_drop)
```

```
✅ Dropped columns: ['Unnamed: 0', 'Unnamed: 0.2', 'Unnamed: 0.1', 'annotation_id', 'annotator', 'author', 'clean_tex
t', 'confidence_score', 'created_at', 'id', 'lead_time', 'predicted_label', 'updated_at']
New shape: (6031, 7)
```

In [ ]:
```
data.columns
```

Out[ ]:
```
Index(['category', 'language', 'likes', 'published_at', 'sentiment',
       'subcategory', 'text'],
      dtype='object')
```

Checking Info

In [ ]:
```
loader.check_info()
```

```
📊 DATA INFO
==================================================
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6031 entries, 0 to 6030
Data columns (total 7 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   category      6031 non-null   object
 1   language      6031 non-null   object
 2   likes         6031 non-null   int64
 3   published_at  6031 non-null   object
 4   sentiment     6031 non-null   object
 5   subcategory   6013 non-null   object
 6   text          6030 non-null   object
dtypes: int64(1), object(6)
memory usage: 329.9+ KB
None

Shape: (6031, 7)
```

Checking for missing Values

In [ ]:
```
loader.check_missing_values()
```

```
🔎 MISSING VALUES
==================================================
category         0
language         0
likes            0
published_at     0
sentiment        0
subcategory     18
text             1
dtype: int64
```

Out[ ]:
```
category         0
language         0
likes            0
published_at     0
sentiment        0
subcategory     18
text             1
dtype: int64
```

Drop missing values

In [ ]:
```
loader.drop_missing_values()
```

```
✅ Dropped 19 rows with missing values.
New shape: (6012, 7)
```

Out[ ]:

	category	language	likes	published_at	sentiment	subcategory	text
0	Irony/Sarcasm	Mixed (Code-switching)	10	2025-07-28T16:36:04Z	Positive	[Irony] Hyperbolic understatement	Napenda hii couple but hapo kwa 29yrs nimekata...
1	Others	Sheng	0	2025-07-28T16:35:44Z	Neutral	[Other] General Others	rada zako wasingam
2	Constructive Criticism	Sheng	0	2025-07-28T16:35:12Z	Positive	[CC] Genuine Praise	Izo chwom mbaya ndio zetuu 😂😂
3	Constructive Criticism	Mixed (Code-switching)	0	2025-07-28T16:35:07Z	Positive	[CC] Genuine Praise	Nka shy 😅
4	Constructive Criticism	Sheng	1	2025-07-28T16:34:43Z	Neutral	[CC] Decision-criticism	Ata kama ni dooh kuna shida zingine siwezi taka
...	...	...	...	...	...	...	...
6026	Irony/Sarcasm	Sheng	0	2025-07-28T16:38:05Z	Negative	[Irony] Mock compliment	Wenye hawajui venye street hu-run mnafloat tu 😂...
6027	Constructive Criticism	English	0	2025-07-28T16:37:50Z	Positive	[CC] Genuine feedback	I can answer that question more further 😂😂😂😂😂😂
6028	Constructive Criticism	English	2	2025-07-28T16:37:39Z	Positive	[CC] Genuine Praise	wedding of the year ❤️❤️
6029	Constructive Criticism	Sheng	1	2025-07-28T16:37:19Z	Neutral	[CC] Supportive Speech	Hiyo wedding nayo , episode 2000 and I can't g...
6030	Constructive Criticism	English	7	2025-07-28T16:36:14Z	Positive	[CC] Genuine feedback	I like the suit Obinna is wearing. Well, the c...

6012 rows × 7 columns

In [ ]:
```
loader.check_shape()
```

```
📐 Current Shape: (6012, 7)
```

Out[ ]:
```
(6012, 7)
```

Check for Duplicates

In [ ]:
```
loader.check_duplicates()
```

```
🔁 Duplicate rows: 1
```

Out[ ]:
```
np.int64(1)
```

In [ ]:
```
loader.drop_duplicates()
```

```
✅ Removed 1 duplicate rows.
New shape: (6011, 7)
```

Out[ ]:

	category	language	likes	published_at	sentiment	subcategory	text
0	Irony/Sarcasm	Mixed (Code-switching)	10	2025-07-28T16:36:04Z	Positive	[Irony] Hyperbolic understatement	Napenda hii couple but hapo kwa 29yrs nimekata...
1	Others	Sheng	0	2025-07-28T16:35:44Z	Neutral	[Other] General Others	rada zako wasingam
2	Constructive Criticism	Sheng	0	2025-07-28T16:35:12Z	Positive	[CC] Genuine Praise	Izo chwom mbaya ndio zetuu 😂😂
3	Constructive Criticism	Mixed (Code-switching)	0	2025-07-28T16:35:07Z	Positive	[CC] Genuine Praise	Nka shy 😅
4	Constructive Criticism	Sheng	1	2025-07-28T16:34:43Z	Neutral	[CC] Decision-criticism	Ata kama ni dooh kuna shida zingine siwezi taka
...	...	...	...	...	...	...	...
6026	Irony/Sarcasm	Sheng	0	2025-07-28T16:38:05Z	Negative	[Irony] Mock compliment	Wenye hawajui venye street hu-run mnafloat tu 😂...
6027	Constructive Criticism	English	0	2025-07-28T16:37:50Z	Positive	[CC] Genuine feedback	I can answer that question more further 😂😂😂😂😂😂
6028	Constructive Criticism	English	2	2025-07-28T16:37:39Z	Positive	[CC] Genuine Praise	wedding of the year ❤️❤️
6029	Constructive Criticism	Sheng	1	2025-07-28T16:37:19Z	Neutral	[CC] Supportive Speech	Hiyo wedding nayo , episode 2000 and I can't g...
6030	Constructive Criticism	English	7	2025-07-28T16:36:14Z	Positive	[CC] Genuine feedback	I like the suit Obinna is wearing. Well, the c...

6011 rows × 7 columns

Standardizing columnns and Column Names

In [ ]:
```
loader.standardize_column_names()
```

```
✅ Column names standardized.
New columns: ['category', 'language', 'likes', 'published_at', 'sentiment', 'subcategory', 'text']
```

Out[ ]:

	category	language	likes	published_at	sentiment	subcategory	text
0	Irony/Sarcasm	Mixed (Code-switching)	10	2025-07-28T16:36:04Z	Positive	[Irony] Hyperbolic understatement	Napenda hii couple but hapo kwa 29yrs nimekata...
1	Others	Sheng	0	2025-07-28T16:35:44Z	Neutral	[Other] General Others	rada zako wasingam
2	Constructive Criticism	Sheng	0	2025-07-28T16:35:12Z	Positive	[CC] Genuine Praise	Izo chwom mbaya ndio zetuu 😂😂
3	Constructive Criticism	Mixed (Code-switching)	0	2025-07-28T16:35:07Z	Positive	[CC] Genuine Praise	Nka shy 😅
4	Constructive Criticism	Sheng	1	2025-07-28T16:34:43Z	Neutral	[CC] Decision-criticism	Ata kama ni dooh kuna shida zingine siwezi taka
...	...	...	...	...	...	...	...
6026	Irony/Sarcasm	Sheng	0	2025-07-28T16:38:05Z	Negative	[Irony] Mock compliment	Wenye hawajui venye street hu-run mnafloat tu😂...
6027	Constructive Criticism	English	0	2025-07-28T16:37:50Z	Positive	[CC] Genuine feedback	I can answer that question more further😂😂😂😂😂😂
6028	Constructive Criticism	English	2	2025-07-28T16:37:39Z	Positive	[CC] Genuine Praise	wedding of the year ❤️❤️
6029	Constructive Criticism	Sheng	1	2025-07-28T16:37:19Z	Neutral	[CC] Supportive Speech	Hiyo wedding nayo , episode 2000 and I can't g...
6030	Constructive Criticism	English	7	2025-07-28T16:36:14Z	Positive	[CC] Genuine feedback	I like the suit Obinna is wearing. Well, the c...

6011 rows × 7 columns

In [ ]:

```
loader.clean_subcategory()
```

```
✅ Cleaned 'subcategory' column.
```

Out[ ]:

	category	language	likes	published_at	sentiment	subcategory	text
0	Irony/Sarcasm	Mixed (Code-switching)	10	2025-07-28T16:36:04Z	Positive	Hyperbolic understatement	Napenda hii couple but hapo kwa 29yrs nimekata...
1	Others	Sheng	0	2025-07-28T16:35:44Z	Neutral	General Others	rada zako wasingam
2	Constructive Criticism	Sheng	0	2025-07-28T16:35:12Z	Positive	Genuine Praise	Izo chwom mbaya ndio zetuu 😂😂
3	Constructive Criticism	Mixed (Code-switching)	0	2025-07-28T16:35:07Z	Positive	Genuine Praise	Nka shy 😅
4	Constructive Criticism	Sheng	1	2025-07-28T16:34:43Z	Neutral	Decision-criticism	Ata kama ni dooh kuna shida zingine siwezi taka
...	...	...	...	...	...	...	...
6026	Irony/Sarcasm	Sheng	0	2025-07-28T16:38:05Z	Negative	Mock compliment	Wenye hawajui venye street hu-run mnafloat tu 😂...
6027	Constructive Criticism	English	0	2025-07-28T16:37:50Z	Positive	Genuine feedback	I can answer that question more further 😂😂😂😂😂😂
6028	Constructive Criticism	English	2	2025-07-28T16:37:39Z	Positive	Genuine Praise	wedding of the year ❤️❤️
6029	Constructive Criticism	Sheng	1	2025-07-28T16:37:19Z	Neutral	Supportive Speech	Hiyo wedding nayo , episode 2000 and I can't g...
6030	Constructive Criticism	English	7	2025-07-28T16:36:14Z	Positive	Genuine feedback	I like the suit Obinna is wearing. Well, the c...

6011 rows × 7 columns

In [ ]:

```
loader.clean_language_column()
```

```
✅ Cleaned 'language' column.
```

Out[ ]:

	category	language	likes	published_at	sentiment	subcategory	text
0	Irony/Sarcasm	Code Switching	10	2025-07-28T16:36:04Z	Positive	Hyperbolic understatement	Napenda hii couple but hapo kwa 29yrs nimekata...
1	Others	Sheng	0	2025-07-28T16:35:44Z	Neutral	General Others	rada zako wasingam
2	Constructive Criticism	Sheng	0	2025-07-28T16:35:12Z	Positive	Genuine Praise	Izo chwom mbaya ndio zetuu 😂😂
3	Constructive Criticism	Code Switching	0	2025-07-28T16:35:07Z	Positive	Genuine Praise	Nka shy 😅
4	Constructive Criticism	Sheng	1	2025-07-28T16:34:43Z	Neutral	Decision-criticism	Ata kama ni dooh kuna shida zingine siwezi taka
...	...	...	...	...	...	...	...
6026	Irony/Sarcasm	Sheng	0	2025-07-28T16:38:05Z	Negative	Mock compliment	Wenye hawajui venye street hu-run mnafloat tu 😂...
6027	Constructive Criticism	English	0	2025-07-28T16:37:50Z	Positive	Genuine feedback	I can answer that question more further 😂😂😂😂😂😂
6028	Constructive Criticism	English	2	2025-07-28T16:37:39Z	Positive	Genuine Praise	wedding of the year ❤️❤️
6029	Constructive Criticism	Sheng	1	2025-07-28T16:37:19Z	Neutral	Supportive Speech	Hiyo wedding nayo , episode 2000 and I can't g...
6030	Constructive Criticism	English	7	2025-07-28T16:36:14Z	Positive	Genuine feedback	I like the suit Obinna is wearing. Well, the c...

6011 rows × 7 columns

📊 Exploratory Data Analysis (EDA)

Introduction

This section explores the structure, distribution, and characteristics of the cyberbullying dataset before modeling.
The goal is to understand patterns in categories, sentiment, text behavior, language usage, and posting activity.

In [ ]:
```
eda = EDA(loader.data)
```

A) Dataset Overview

The dataset overview examined the structure and quality of the data.
This included checking the number of rows and columns, data types, column names, and missing values.

This step ensured the dataset is clean, properly formatted, and ready for deeper exploratory analysis and modeling.

In [ ]:
```
eda.dataset_overview()
```

```
📊 DATASET OVERVIEW
============================================================
Shape: (6011, 7)

Columns:
['category', 'language', 'likes', 'published_at', 'sentiment', 'subcategory', 'text']

Data Types:
category        object
language        object
likes            int64
published_at    object
sentiment       object
subcategory     object
text            object
dtype: object

Missing Values:
category        0
language        0
likes           0
published_at    0
sentiment       0
subcategory     0
text            0
dtype: int64
```

B) Target Distribution Analysis

This analysis explored the distribution of posts across different cyberbullying categories and sentiment classes.

Understanding class balance is important for machine learning, as imbalanced data may require resampling or class weighting. The sentiment distribution also provides insight into the overall emotional tone of the dataset.

In [ ]:
```
eda.target_distribution("category")
```

```
Class Percentages:
category
Constructive Criticism    52.74
Others                    21.73
Cyberbullying             10.66
Irony/Sarcasm              8.53
Offensive                  4.62
Misinformation             1.10
Hate Speech                0.62
Name: proportion, dtype: float64
```

In [ ]:
```
eda.target_distribution("sentiment")
```

```
Class Percentages:
sentiment
Neutral     38.20
Positive    33.72
Negative    28.08
Name: proportion, dtype: float64
```

C) Subcategory Analysis

The subcategory analysis examined more detailed classifications of cyberbullying behavior.

This helps identify dominant bullying patterns and determines whether fine-grained classification is feasible.
It also highlights subcategories that may require balancing during model training.

In [ ]:
```
eda.subcategory_analysis()
```

D) Text Length Analysis

Text length analysis evaluated the distribution of post lengths in characters.

This insight helps determine appropriate tokenization limits, sequence lengths for NLP models, and detection of unusually short or long posts that may affect model performance.

In [ ]:
```
eda.text_length_analysis()
```

```
Text Length Summary:
count    6011.000000
mean       49.085676
std        49.553109
min         1.000000
25%        21.000000
50%        36.000000
75%        61.000000
max      1020.000000
Name: text_length, dtype: float64
```

E) Language Distribution

The language distribution analysis identified the dominant languages used in the dataset.

This is important for preprocessing decisions such as stopword removal, tokenization strategy, and determining whether multilingual modeling is required.

In [ ]:
```
eda.language_insights()
```

```
Language Percentages:
language
Code Switching    38.58
English           24.07
Sheng             18.28
Swahili           16.54
Plain Emoji        2.53
Name: proportion, dtype: float64
```

F) Peak Posting Time Analysis (Kenyan Time - EAT)

This analysis converted timestamps to Kenyan time (EAT - UTC+3) and examined hourly posting patterns.

Identifying peak activity hours provides behavioral insights into when cyberbullying is most active and may support real-time moderation strategies.
Weekend and weekday comparisons further contextualize user behavior.

In [ ]:
```
eda.peak_posting_times("published_at")
```

```
🕒 Posting Activity by Hour
============================================================
hour
0     478
1     179
2     100
3      77
4      63
5      48
6      62
7     100
8     186
9     218
10    230
11    289
12    281
13    307
14    225
15    219
16    237
17    218
18    222
19    365
20    454
21    490
22    396
23    567
Name: count, dtype: int64

🔥 Peak Posting Hour:
23:00

📅 Weekend vs Weekday Activity
============================================================
is_weekend
False    4228
True     1783
Name: hour, dtype: int64
```

Out[ ]:
```
hour
0     478
1     179
2     100
3      77
4      63
5      48
6      62
7     100
8     186
9     218
10    230
11    289
12    281
13    307
14    225
15    219
16    237
17    218
18    222
19    365
20    454
21    490
22    396
23    567
Name: count, dtype: int64
```

G) Key Findings Summary

The key findings section summarized major insights from the EDA process, including dominant categories, sentiment trends, language patterns, and average text length.

These insights guide preprocessing decisions, feature engineering, and model development strategies for the cyberbullying detection system.

In [ ]:
```
eda.key_findings()
```

```
🔎 KEY FINDINGS SUMMARY
============================================================
Total Samples: 6011

Top Category:
Constructive Criticism

Top Sentiment:
Neutral

Most Common Language:
Code Switching

Average Text Length: 49.09 characters
```

PRE-PROCESSING

In [ ]:
```
from inference.preprocesssing import TextPreprocessor
preprocessor = TextPreprocessor()
final_df = preprocessor.transform(data, column="text")
final_df.head()
```

Out[ ]:

	category	language	likes	published_at	sentiment	subcategory	text	clean_text
0	Irony/Sarcasm	Code Switching	10	2025-07-28T16:36:04Z	Positive	Hyperbolic understatement	Napenda hii couple but hapo kwa 29yrs nimekata...	napenda hii couple but hapo kwa 29yrs nimekata...
1	Others	Sheng	0	2025-07-28T16:35:44Z	Neutral	General Others	rada zako wasingam	rada zako wasingam
2	Constructive Criticism	Sheng	0	2025-07-28T16:35:12Z	Positive	Genuine Praise	Izo chwom mbaya ndio zetuu 😂 😂	izo chwom mbaya ndio zetuu face_with_tears_of_...
3	Constructive Criticism	Code Switching	0	2025-07-28T16:35:07Z	Positive	Genuine Praise	Nka shy 😅	nka shy grinning_face_with_sweat
4	Constructive Criticism	Sheng	1	2025-07-28T16:34:43Z	Neutral	Decision-criticism	Ata kama ni dooh kuna shida zingine siwezi taka	ata kama ni dooh kuna shida zingine siwezi taka

Feature Engineering

In [ ]:

```
from inference.Feature_engineering import FeatureEngineer

fe = FeatureEngineer()

X_train_text, X_test_text, y_train, y_test = fe.split(
    final_df["clean_text"],
    final_df["language"]
)

X_train = fe.fit_transform(X_train_text)
X_test = fe.transform(X_test_text)
```

MODELLING

1. Language Detection Model

The goal of this model is to identify the dominant language of each comment.

```
* English

*Swahili

* Kenyan Slang.
```

The kenyan social media content is highly code-mixed.

APPROACH

A linear Support vector classifier(LinearSVC) was used due to :

- Its strong performance on text classification
- Its efficient with high dimentional sparse features
- Robust to noisy social media text.

A custom Feature Engineering was used to perform:

- TF-IDF vectorization
- Capture character level patterns
- Handle train test_splitting

2. Sentiment Analysis Model

The goal of this model is to classify the emotional tone of each comment. Sentiment falls under three major categories.

1. Positive
2. Neutral
3. Negative

Understanding the sentiment of a comment help identify potentially harmful comments and guide on further classification

Approach

A linear support vector classifier(LinearSVC) was used due to its capabilities to;

- Efficiently classify texts
- Strong performance on sparse TF-IDF features. Features include:
- Word level TF-IDF representation
- Sentiment lexicon to capture emotional tone.

Model performance

The model was evaluated using

- Accuracy
- Precision
- Recall
- F1-score

3. Category Classification Model

The objective of the classification model is to clasify models into high level categories:

1. Constructive
2. Harmful
3. Offensive
4. Cyberbullying
5. Others
6. Irony/Sarcasm.

Approach

A high level transformer- model based(XLM-RoBERTa) was used, it has the ability to;

- Capture contexual meaning
- Handle multilingual text
- understand sarcasm and nuanced expressions

The model was fine-tuned on labeled data using tokenized text inputs.

Only negative and neutral comments were used.(this improves potentially harmful content)

4. Subcategory Classification Model

The goal of the model is to fine tune the classification within each major category. Example: Cyberbulling category hold the followin subcategories

- Direct insult
- Threat
- Exclusion tactics
- Sexual harassment
- Trolling
- Defamation
- Body Shaming
- Humiliation The same subcategory classification happens across all other categories.

Approach

Each category was trained on a different transformer model
The model focuses only on its respective subset of data.
To fine-tune each category the (XLM-RoBERTa transformer was used.)

MODEL EVALUATION

The goal of this phase is to assess the performance of the hierachical models. The models were assesed using the standaed classification metrics to measure their effectiveness. The following metrics were used:

- **Accuracy** → Overall correctness of predictions
- **Precision** → Proportion of correct positive predictions
- **Recall** → Ability to identify all relevant instances
- **F1-score** → Balance between precision and recall

The performance of the model was not good as the models achieved accuracy of about 53% which is very low.
An accuracy of 50% indicates that the models could accurately predict 50% of the data set correctly.

DEPLOYMENT

6.1 System Architecture

The deployed system follows a modular architecture where each component performs a specific task in the pipeline.

Workflow

User Input
↓
Text Preprocessing
↓
Language Detection
↓
Sentiment Analysis
↓
Category Classification
↓
Subcategory Classification

↓
Final Output

This design ensures scalability, modularity, and efficient processing of multilingual data.

6.2 Model Hosting

The trained models were hosted on Hugging Face Hub to enable scalable and efficient deployment.

Advantages

- Reduces local storage requirements
- Enables easy model version control
- Allows dynamic loading of models during application runtime
- Supports cloud-based deployment

The application downloads and loads the models when initialized, ensuring flexibility and portability.

User Interface

An interactive web application was developed using Streamlit to allow users to analyze comments in real time.

The interface provides a simple and intuitive way to interact with the machine learning pipeline.

Single Comment Analysis

Users can input a single comment into the system.

The application returns:

- Detected language
- Sentiment classification
- Category
- Subcategory

- Toxicity level
- Explanation of the classification

This allows real-time moderation of individual comments.

Batch Comment Analysis

The system supports batch processing through CSV file uploads.

Users can upload multiple comments, and the system processes each comment through the pipeline.

Output

- Structured table of predictions
- Language distribution
- Sentiment distribution
- Category analysis
- Subcategory This enables efficient moderation of large volumes of data.

6.4 Model Optimization

Several strategies were implemented to improve performance:

- **Model caching** to prevent reloading during runtime
- **Hierarchical classification** to reduce computational load
- **External model hosting** to improve scalability

These optimizations ensure faster and more efficient predictions.

Future Improvements

- Deploy the system as a REST API for integration with external platforms
- Implement GPU-based inference for faster transformer predictions

- Expand support to additional platforms such as TikTok and Twitter
- Continuously update models with new slang and evolving cyberbullying patterns

Conclusion

The system successfully transforms machine learning models into a practical moderation tool capable of detecting cyberbullying in multilingual Kenyan social media content.

The use of a hierarchical pipeline and cloud-based model hosting ensures scalability, efficiency, and real-world applicability.